Mixture & Ratio – Practice Questions

All questions skipped — attempt before reviewing solutions.

Question 1

A certain amount of water was poured into a 300 litre container and the remaining portion of the container was filled with milk. Then an amount of this solution was taken out from the container which was twice the volume of water that was earlier poured into it, and water was poured to refill the container again. If the resulting solution contains 72% milk, then the amount of water, in litres, that was initially poured into the container was

(Type in the Answer — numerical answer required)

Question 2

Anil mixes cocoa with sugar in the ratio 3:2 to prepare mixture A, and coffee with sugar in the ratio 7:3 to prepare mixture B. He combines mixtures A and B in the ratio 2:3 to make a new mixture C. If he mixes C with an equal amount of milk to make a drink, then the percentage of sugar in this drink will be

A) 17
B) 16
C) 21
D) 24

Question 3

The scores of Amal and Bimal in an examination are in the ratio 11 : 14. After an appeal, their scores increase by the same amount and their new scores are in the ratio 47 : 56. The ratio of Bimal's new score to that of his original score is

A) 4 : 3
B) 8 : 5
C) 5 : 4
D) 3 : 2

Question 4

A 20% ethanol solution is mixed with another ethanol solution, say S of unknown concentration in the proportion 1:3 by volume. This mixture is then mixed with an equal volume of 20% ethanol solution. If the resultant mixture is a 31.25% ethanol solution, then the unknown concentration of S is

A) 30%
B) 40%
C) 50%
D) 60%

Question 5

A jar contains a mixture of 175 ml water and 700 ml alcohol. Gopal takes out 10% of the mixture and substitutes it by water of the same amount. The process is repeated once again. The percentage of water in the mixture is now

A) 30.3
B) 35.2
C) 25.4
D) 20.5

Question 6

There are two drums, each containing a mixture of paints A and B. In drum 1, A and B are in the ratio 18 : 7. The mixtures from drums 1 and 2 are mixed in the ratio 3 : 4 and in this final mixture, A and B are in the ratio 13 : 7. In drum 2, then A and B were in the ratio

A) 251 : 163
B) 239 : 161
C) 220 : 149
D) 229 : 141

Question 7

The strength of an indigo solution in percentage is equal to the amount of indigo in grams per 100 cc of water. Two 800 cc bottles are filled with indigo solutions of strengths 33% and 17%, respectively. A part of the solution from the first bottle is thrown away and replaced by an equal volume of the solution from the second bottle. If the strength of the indigo solution in the first bottle has now changed to 21% then the volume, in cc, of the solution left in the second bottle is

(Type in the Answer — numerical answer required)

Question 8

A glass contains 500 cc of milk and a cup contains 500 cc of water. From the glass, 150 cc of milk is transferred to the cup and mixed thoroughly. Next, 150 cc of this mixture is transferred from the cup to the glass. Now, the amount of water in the glass and the amount of milk in the cup are in the ratio

A) 1:01
B) 10:13
C) 3:10
D) 10:03

Question 9

A container has 40 liters of milk. Then, 4 liters are removed from the container and replaced with 4 liters of water. This process of replacing 4 liters of the liquid in the container with an equal volume of water is continued repeatedly. The smallest number of times of doing this process, after which the volume of milk in the container becomes less than that of water, is

(Type in the Answer — numerical answer required)

Question 10

A solution, of volume 40 litres, has dye and water in the proportion 2 : 3. Water is added to the solution to change this proportion to 2 : 5. If one-fourth of this diluted solution is taken out, how many litres of dye must be added to the remaining solution to bring the proportion back to 2 : 3?

(Type in the Answer — numerical answer required)

Question 11

An alloy is prepared by mixing three metals A, B and C in the proportion 3 : 4 : 7 by volume. Weights of the same volume of the metals A, B and C are in the ratio 5 : 2 : 6. In 130 kg of the alloy, the weight, in kg, of the metal C is:

A) 48
B) 84
C) 70
D) 96

Question 12

Rajesh and Vimal own 20 hectares and 30 hectares of agricultural land, respectively, which are entirely covered by wheat and mustard crops. The cultivation area of wheat and mustard in the land owned by Vimal are in the ratio 5 : 3. If the total cultivation area of wheat and mustard are in the ratio 11 : 9, then the ratio of cultivation area of wheat and mustard in the land owned by Rajesh is

A) 4:3
B) 7:9
C) 3:7
D) 1:1

Question 13

A vessel contained a solution of acid and water. When 2 litres of water were added, the new solution had a 50% acid concentration. When 15 litres of acid were then added, the final solution had an 80% acid concentration. What was the ratio of water to acid in the original solution?

A) 5 : 3
B) 3 : 5
C) 5 : 4
D) 4 : 5

Question 14

When Rajesh's age was the same as the present age of Garima, the ratio of their ages (Rajesh's to Garima's) was 3 : 2. When Garima's age becomes the same as the present age of Rajesh, the ratio of the ages of Rajesh and Garima will become:

A) 3 : 2
B) 4 : 3
C) 5 : 4
D) 2 : 1

Question 15

There are two containers of the same volume, first container half-filled with sugar syrup and the second container half-filled with milk. Half the content of the first container is transferred to the second container, and then the half of this mixture is transferred back to the first container. Next, half the content of the first container is transferred back to the second container. Then the ratio of sugar syrup and milk in the second container is

A) 4:5
B) 6:5
C) 5:4
D) 5:6

Question 16

From a container filled with milk, 9 litres of milk are drawn and replaced with water. Next, from the same container, 9 litres are drawn and again replaced with water. If the volumes of milk and water in the container are now in the ratio of 16:9, then the capacity of the container, in litres, is:

(Type in the Answer — numerical answer required)

Question 17

If a certain weight of an alloy of silver and copper is mixed with 3 kg of pure silver, the resulting alloy will have 90% silver by weight. If the same weight of the initial alloy is mixed with 2 kg of another alloy which has 90% silver by weight, the resulting alloy will have 84% silver by weight. Then, the weight of the initial alloy, in kg, is

A) 3.5
B) 2.5
C) 3
D) 4

Question 18

In a village, the ratio of the number of males to females is 5 : 4. The ratio of the number of literate males to literate females is 2 : 3. The ratio of the number of illiterate males to illiterate females is 4 : 3. If 3600 males in the village are literate, then the total number of females in the village is:

(Type in the Answer — numerical answer required)

Question 19

Amal buys 110 kg of syrup and 120 kg of juice, syrup being 20% less costly than juice, per kg. He sells 10 kg of syrup at 10% profit and 20 kg of juice at 20% profit. Mixing the remaining juice and syrup, Amal sells the mixture at ■308.32 per kg and makes an overall profit of 64%. Then, Amal's cost price for syrup, in rupees per kg, is:

(Type in the Answer — numerical answer required)

Question 20

A mixture contains lemon juice and sugar syrup in equal proportion. If a new mixture is created by adding this mixture and sugar syrup in the ratio 1 : 3, then the ratio of lemon juice and sugar syrup in the new mixture is:

A) 1 : 4

B) 1 : 5

C) 1 : 6

D) 1 : 7